Example 1

- Every cadet runs a timed sprint course as part of camp trials.
- The coaches want us to determine if the Thunder Squadron and Lightning Squadron are finishing the sprint course at different times.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = sprint_time_sec,
                                     grouping = squadron,
                                     alternative = "two",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 44.4643
x̄[Thunder Squadron] = 41.8188
Point estimate: x̄[Lightning Squadron] − x̄[Thunder Squadron] = 2.6455

95% CI for μ[Lightning Squadron] − μ[Thunder Squadron]: (0.3719, 4.9192)

Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] = 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] ≠ 0
Test statistic: t(27.81) = 2.384
p-value: 0.024
Conclusion: Reject the null hypothesis (p = 0.024 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 2

- At camp, judges score each cadet's aerial loop maneuver on a 0–100 accuracy scale during trials.
- Based on her gut feeling, the head judge believes that, on average, the Thunder Squadron is performing more accurate loop maneuvers.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = loop_accuracy_pct,
                                     grouping = squadron,
                                     alternative = "less",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 78.4071
x̄[Thunder Squadron] = 81.2875
Point estimate: x̄[Lightning Squadron] − x̄[Thunder Squadron] = -2.8804

95% CI for μ[Lightning Squadron] − μ[Thunder Squadron]: (-7.606, 1.8453)

Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] ≥ 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] < 0
Test statistic: t(25.18) = -1.255
p-value: 0.111
Conclusion: Fail to reject the null hypothesis (p = 0.111 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Example 3

- Each cadet's aerial accuracy is scored when they arrive at camp, and scored again at the end after two weeks of drills.
- Coaches want to know whether cadets are actually flying more accurately by the end of camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = accuracy_score_post,
                                   col2 = accuracy_score_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[accuracy_score_post] - x̄[accuracy_score_pre] = 6.7267
SD of differences: 5.1583

95% CI for μ[accuracy_score_post] - μ[accuracy_score_pre]: (4.8005, 8.6528)

Hypotheses:
H₀: μ[accuracy_score_post] - μ[accuracy_score_pre] ≤ 0
H₁: μ[accuracy_score_post] - μ[accuracy_score_pre] > 0
Test statistic: t(29) = 7.143
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**

Example 4

- Each cadet's recovery time after a hard trial is timed at the start of camp and again at the end.
- Coaches want to know whether cadets are recovering faster by the end of camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = recovery_time_min_post,
                                   col2 = recovery_time_min_pre,
                                   alternative = "less",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[recovery_time_min_post] - x̄[recovery_time_min_pre] = -3.26
SD of differences: 3.3987

95% CI for μ[recovery_time_min_post] - μ[recovery_time_min_pre]: (-4.5291,
-1.9909)

Hypotheses:
H₀: μ[recovery_time_min_post] - μ[recovery_time_min_pre] ≥ 0
H₁: μ[recovery_time_min_post] - μ[recovery_time_min_pre] < 0
Test statistic: t(29) = -5.254
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 1

- Spectators toss bits to cadets who pull off a standout run during trials.
- Camp organizers want to know if the squadrons are earning prize bits equally.

```
wonderbolt_camp |> independent_means(outcome = prize_bits_won,
                                     grouping = squadron,
                                     alternative = "two",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 27.7857
x̄[Thunder Squadron] = 36.1875
Point estimate: x̄[Lightning Squadron] − x̄[Thunder Squadron] = -8.4018

95% CI for μ[Lightning Squadron] − μ[Thunder Squadron]: (-27.9469, 11.1433)

Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] = 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] ≠ 0
Test statistic: t(26.86) = -0.882
p-value: 0.385
Conclusion: Fail to reject the null hypothesis (p = 0.385 ≥ α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 2

- Spectators tip cadets with bits after their runs. The amount tipped is recorded once early in camp and once again near the end.
- Camp organizers want to know whether cadets are earning more in tips as they improve over camp.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = tip_bits_post,
                                   col2 = tip_bits_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[tip_bits_post] - x̄[tip_bits_pre] = 17.9333
SD of differences: 13.4982

95% CI for μ[tip_bits_post] - μ[tip_bits_pre]: (12.893, 22.9737)

Hypotheses:
H₀: μ[tip_bits_post] - μ[tip_bits_pre] ≤ 0
H₁: μ[tip_bits_post] - μ[tip_bits_pre] > 0
Test statistic: t(29) = 7.277
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 3

- A sound crew records the peak crowd noise (in decibels) during each cadet's run.
- Camp organizers believe that the Thunder Squadron is inspiring a higher level of noise.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = cheer_decibels,
                                     grouping = squadron,
                                     alternative = "less",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 84.9
x̄[Thunder Squadron] = 90.8187
Point estimate: x̄[Lightning Squadron] − x̄[Thunder Squadron] = -5.9187

95% CI for μ[Lightning Squadron] − μ[Thunder Squadron]: (-8.6025, -3.235)

Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] ≥ 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] < 0
Test statistic: t(21.92) = -4.575
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 4

- The camp medic tracks how many days each cadet spends recovering from an injury over the two-week camp.
- Coaches want to know if the two training styles lead to the same number of injury days.
- Using the `independent_means()` function,

```
wonderbolt_camp |> independent_means(outcome = injury_days,
                                     grouping = squadron,
                                     alternative = "two",
                                     mu = 0,
                                     alpha = 0.05)
```

```
Independent means (Welch's t-test)

x̄[Lightning Squadron] = 1.7143
x̄[Thunder Squadron] = 1.3125
Point estimate: x̄[Lightning Squadron] − x̄[Thunder Squadron] = 0.4018

95% CI for μ[Lightning Squadron] − μ[Thunder Squadron]: (-0.515, 1.3186)

Hypotheses:
H₀: μ[Lightning Squadron] − μ[Thunder Squadron] = 0
H₁: μ[Lightning Squadron] − μ[Thunder Squadron] ≠ 0
Test statistic: t(25.11) = 0.902
p-value: 0.375
Conclusion: Fail to reject the null hypothesis (p = 0.375 ≥ α = 0.050)
```

- **Hypotheses:**

- **Test Statistic and *p*-Value**

- **Rejection Region**

- **Conclusion and interpretation**

Let's Practice! 5

- Each cadet's wing-flap rate while hovering in place is measured on arrival, then measured again after two weeks of drills.
- Coaches want to know whether training actually changes a cadet's wing-flap rate.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = wing_flap_rate_post,
                                   col2 = wing_flap_rate_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[wing_flap_rate_post] - x̄[wing_flap_rate_pre] = 3.7833
SD of differences: 4.7192

95% CI for μ[wing_flap_rate_post] - μ[wing_flap_rate_pre]: (2.0212, 5.5455)

Hypotheses:
H₀: μ[wing_flap_rate_post] - μ[wing_flap_rate_pre] ≤ 0
H₁: μ[wing_flap_rate_post] - μ[wing_flap_rate_pre] > 0
Test statistic: t(29) = 4.391
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**

Let's Practice! 6

- Each cadet rates how confident they feel about their flying on a self-reported scale, once at the start of camp and once at the end.
- Camp staff want to know whether two weeks of training changes how confident cadets feel.
- Using the `dependent_means()` function,

```
wonderbolt_camp |> dependent_means(col1 = confidence_rating_post,
                                   col2 = confidence_rating_pre,
                                   alternative = "greater",
                                   mu = 0,
                                   alpha = 0.05)
```

```
Paired (dependent) means

Complete pairs: 30
Point estimate: x̄[confidence_rating_post] - x̄[confidence_rating_pre] = 1.0667
SD of differences: 1.3629

95% CI for μ[confidence_rating_post] - μ[confidence_rating_pre]: (0.5578,
1.5756)

Hypotheses:
H₀: μ[confidence_rating_post] - μ[confidence_rating_pre] ≤ 0
H₁: μ[confidence_rating_post] - μ[confidence_rating_pre] > 0
Test statistic: t(29) = 4.287
p-value: < 0.001
Conclusion: Reject the null hypothesis (p = < 0.001 < α = 0.050)
```

- **Hypotheses:**
- **Test Statistic and *p*-Value**
- **Rejection Region**
- **Conclusion and interpretation**